HOLY ANGEL UNIVERSITY

SMART QUEUE MANAGEMENT SYSTEM FOR UNIVERSITY CLINICS

A Capstone Project
Presented to the Faculty of the
School of Computing
Holy Angel University

In Partial Fulfillment
of the Requirements for the Degree
Bachelor of Science in Information Technology with Area of Specialization in Web Development

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Proposal manuscript — draft (Introduction to Conceptual Framework). Later sections follow in the complete proposal. Dummy document generated for testing the HAU-SOC Thesis & Capstone prototype. All names, data and references are placeholders.

SCHOOL OF COMPUTING

APPROVAL SHEET

This capstone entitled **"SMART QUEUE MANAGEMENT SYSTEM FOR UNIVERSITY CLINICS"**, prepared and submitted in partial fulfillment of the requirements for the degree Bachelor of Science in Information Technology with Area of Specialization in Web Development has been examined and is recommended for acceptance and oral examination.

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ORAL EXAMINATION

Approved by the committee of oral examiners on
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APPROVAL

Accepted and approved in partial fulfillment of the requirements for the degree Bachelor of Science in Information Technology with Area of Specialization in Web Development.

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Smart Queue Management System for University Clinics

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Holy Angel University
Information Technology

Abstract
The abstract is written after the study is completed and is left blank at the proposal stage.

SMART QUEUE MANAGEMENT SYSTEM FOR UNIVERSITY CLINICS

Introduction

University clinics serve a steady stream of students, faculty and staff who arrive for consultations, medical clearances, dental check-ups and the release of laboratory results. At most campuses the arrival of these patients is still managed by a paper logbook at the reception desk and a verbal call of names once a nurse or physician is free.

Digital queue management systems address this by issuing a numbered ticket, displaying the current number, and notifying the patient when their turn approaches. Commercial systems are priced and designed for large institutions; a lightweight web-based system running on the phones students already carry would suit a campus clinic better.

The problem this study addresses is the following: patients cannot see their position in line, staff are interrupted by repeated questions, urgent cases wait behind routine clearances, and the clinic cannot report service volume. This capstone project therefore proposes the Smart Queue system.

Review of Related Literature

This section reviews studies and existing systems related to the Smart Queue system and the methods used to evaluate it.

Queueing Theory and Perceived Waiting Time

For a single-server clinic desk, the expected waiting time grows sharply as utilization approaches one (Example & Writer, 2022). Uncertain waits feel longer than known, finite waits of the same duration (Placeholder, 2021), which supports giving every patient a live view of the queue.

Web-Based and Mobile Queue Management Systems

A registrar queue built as a progressive web application reduced counter congestion by letting students wait outside and return when notified (Author et al., 2023). The proposed system adopts ticket issuance without a kiosk, a public display and staff call controls.

Notification Channels and Response Behavior

Email and push notifications are nearly free to send, while SMS carries per-message costs small institutions struggle to sustain (Writer, 2022).

Software Quality Evaluation

Capstone projects in the Philippines commonly ask IT experts to rate a system against ISO/IEC 25010 characteristics on a Likert scale (Example, 2023).

Synthesis

The reviewed studies show that the problem is common and that a focused, web-based solution can address it at low cost. None of the systems reviewed fits the setting of this study exactly, which is the gap the Smart Queue system fills.

Conceptual Framework

The study follows the Input–Process–Output model shown in Figure 1.

Figure 1

Conceptual Framework of the Study

INPUT	PROCESS	OUTPUT
Current clinic process Staff and patient requirements React, Firebase	Requirement analysis Design and development (Scrum) Testing and evaluation	Smart Queue Management System for University Clinics Evaluation results

Draft: Objectives of the Study, Scope and Delimitations and Method are being written and will be in the complete proposal manuscript.